

Justin Jin

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EDUCATION

Arizona State University

Master of Science - Computer Science

Bachelor of Science - Computer Science | Minor - Mathematics

Tempe, AZ

GPA 3.82

July 2025

GPA 3.87

May 2024

SKILLS

Languages: C#, Java, Python, TypeScript, JavaScript, C++, Go, SQL, HTML/CSS, YAML, Bash

Frameworks: React, Angular, Node.js, Express.js, Spring, Flask

Database: PostgreSQL, MySQL, MongoDB, Redis

Dev. Tools: Linux, Git, AWS, GCP, Docker, Kubernetes, GitLab CI/CD, Terraform, Helm, Swagger, Copilot

EXPERIENCE

Research Assistant

Jan 2025 - Jul 2025

J.Liang Lab @ ASU

Tempe, AZ

- Built scalable Python data pipelines and AI models for large medical imaging datasets on Linux GPU clusters.
- Extended a PyTorch multi-task model to support metadata-driven learning, increasing avg. performance by ~10%.
- Reduced preprocessing and validation runtime by 30% per epoch (40 min → 30 min) through optimized parallel data loading and efficient augmentation
- Built evaluation and reproducibility pipeline with automated logging, metrics (Dice, FROC), and reporting via WandB.
- Automated batch experiments using shell scripts and SLURM scheduler in a distributed HPC environment.

Software Engineer Intern

Aug 2023 - May 2024

AndBounds

Scottsdale, AZ

- Re-architected an ASP.NET Core monolith into 18 gRPC microservices, containerized with Docker and deployed to Kubernetes on GCP with rolling updates and HPA, enabling independent zero-downtime service deployments.
- Implemented Tyk API Gateway and Istio service mesh to provide load balancing, rate limiting, and secure service-to-service communication (mTLS).
- Enhanced React homepage with scroll-driven animations and modern UI interactions.
- Built GitLab CI/CD pipelines with xUnit test automation, Helm-based Kubernetes deployments, and Terraform infrastructure provisioning; reduced release time from 1 hour manual process to <10 minutes automated.
- Leveraged GitHub Copilot (GenAI) to accelerate service decomposition and boilerplate generation, contributing to the delivery of the migrated platform within 6 months.

RELEVANT PROJECTS

CanvasLMS Web-Embed Attendance Tool

May 2023 - Dec 2023

ASU CS Dept. TA Project

Tempe, AZ

- Developed a Spring Boot service to ingest door-scanner events and automatically sync attendance to Canvas LMS.
- Designed RESTful APIs with request validation, structured error handling, and JWT-based service authentication; deployed on AWS EC2.
- Implemented OAuth 2.0 flow for secure Canvas LMS API access and automated grade updates.
- Built Angular UI and integrated PostgreSQL data persistence using JPA/Hibernate.

Health Insurance 4 Dummies - AI Insurance Cost Predictor

Oct 2023

Cal Hack 10.0 Hackathon - 2nd Pl.

San Francisco, CA

- Built and deployed a full-stack AI web application (Flask, React, Tailwind, AWS EC2).
- Built an FNN-based healthcare cost prediction model trained on public insurance risk datasets.
- Integrated InterSystems GPU infrastructure (L40) for scalable training and real-time predictions.
- Generated automated analytical reports using TogetherAI LLM APIs and custom prompt engineering.

Liquid-Propulsion Rocket Flight Controller Firmware

Aug 2022 - May 2023

Avionics - Sun Devil Rocketry Club

Tempe, AZ

- Developed embedded firmware for custom ARM Cortex-M7 microcontroller in C using HAL, including sensor and external flash drivers; enabled reliable data collection for two-stage rockets across 2 launches up to 41,000 ft.
- Reduced CPU blocking 13× (1.3 ms → 0.1 ms) by implementing non-blocking DMA.